

HOSPITAL MANAGEMENT SYSTEM MODIFICATION PROMPT (PART 2)

Continue modifying the existing Hospital Management System. Do not rebuild it from scratch. All existing modules must remain functional while implementing the following improvements. Every module must be connected with the patient's unique profile and Medical Record (EMR).

5. LABORATORY MODULE

Laboratory Dashboard

Redesign the Laboratory Dashboard to make it more practical.

Dashboard should include:

- Pending Laboratory Requests
- Accepted Requests
- Sample Collection Queue
- Testing in Progress
- Completed Reports
- Search Patient
- QR Code Scanner
- Recent Uploaded Reports

Large Search Box

Search using:

- Patient ID
- QR Code
- Phone Number
- Name
- Registration Number

Laboratory Workflow

Doctor requests laboratory investigation.

↓

Laboratory immediately receives the request.



Laboratory staff searches the patient or opens the pending request.



Laboratory staff reviews:

- Doctor Name
- Department
- Requested Tests
- Clinical Note
- Uploaded Referral Letter (PDF/Scanned)
- Priority
- Request Date & Time

Accept Request

Provide an **Accept Request** button.

This is only to acknowledge the request inside the system.

If the patient arrives with a physical referral instead of an electronic request, laboratory staff must also be able to manually create a laboratory record after searching the patient.

Both workflows must be supported.

Laboratory Permissions

Laboratory staff CAN:

- Search patients
- View patient information
- View doctor's referral
- Upload laboratory report
- Upload scanned report
- Upload PDF
- Write laboratory note
- Update report status

Laboratory staff CANNOT:

- Print patient medical history
- Edit doctor notes
- Delete medical records

- Access unrelated financial modules
 - Modify billing history
-

Laboratory Payment

Laboratory staff should **not** directly manage billing.

Payment can be completed by:

- Cashier
- Laboratory Counter (only for laboratory services)

Laboratory billing must not include admission, pharmacy, OT, or radiology payments.

Cashier can collect payment for **all departments**.

Laboratory Report Upload

Each laboratory investigation must support:

- Upload PDF
- Upload scanned report
- Write report summary
- Add comments
- Upload multiple files

All uploaded files must automatically appear in:

- Patient Medical History
 - Doctor Dashboard
 - Patient Portal
 - Super Admin
 - Nursing Dashboard (view only)
 - Operation Theatre (view only when required)
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Laboratory Timeline

Each uploaded report should automatically record:

- Laboratory Name
- Technician Name
- Upload Date

- Upload Time
- Test Name
- Report Status
- File Type

Everything must become part of the patient's permanent Medical Record.

6. RADIOLOGY MODULE

Radiology should work almost exactly like Laboratory.

Dashboard includes:

- Pending Radiology Requests
- Accepted Requests
- Imaging Queue
- Completed Reports
- Search Patient
- QR Scanner

Search using:

- Patient ID
 - QR
 - Phone
 - Name
-

Radiology Request

Doctor creates radiology request.

↓

Radiology receives request.

↓

Radiology reviews:

- Requested Imaging
- Clinical Notes
- Priority
- Uploaded Referral Letter

Upload Results

Support:

- PDF Upload
- Scanned Report
- Image Upload (X-ray, CT, MRI, Ultrasound, etc.)
- Radiologist Note

Automatically sync to:

- Medical History
- Doctor Dashboard
- Patient Portal
- Super Admin
- Nursing Dashboard
- Operation Theatre

7. CASHIER MODULE

Cashier is the central payment authority.

Cashier can collect payment for:

- OPD
- Registration
- Laboratory
- Radiology
- Admission
- Operation Theatre
- Blood Bank
- Pharmacy
- Other hospital services

Cashier cannot modify clinical records.

Payment Methods

Every billing screen should support:

- Cash (Default)
- eSewa

- Card
 - Insurance
 - Bank Transfer
 - Other
-

Receipt Format

Every payment receipt must have the same professional layout.

Include:

- Hospital Logo
 - Patient Name
 - Phone Number (Compulsory)
 - Patient ID
 - Department
 - Bill Number
 - QR Code
 - Unique Barcode
 - Counter Name
 - Counter Staff Name
 - Payment Method
 - Date
 - Time
 - Total Amount
 - Discount
 - Insurance Coverage
 - Final Amount Paid
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8. MEDICAL RECORDS (EMR)

Medical Records should become the central hub of the entire system.

Current implementation is insufficient and must be redesigned.

Search Patient

Search using:

- Patient ID
- QR Code
- Phone Number

- Name
-

Open Full Medical Record

Display a complete patient timeline organized by department.

Examples:

- OPD Visits
- Admission History
- Laboratory Reports
- Radiology Reports
- Pharmacy Records
- Blood Bank Records
- Operation Theatre Records
- Nursing Notes
- Doctor Notes
- Uploaded PDFs
- Scanned Documents

Each section should be expandable.

Upload Options

Every authorized department should have:

- Write Note
- Upload PDF
- Upload Scanned File
- Save to Medical History

No manual duplication should be required.

Once uploaded, the document automatically becomes part of the patient's permanent Medical Record.

Medical Record Permissions

Super Admin

Can:

- View
- Edit
- Delete
- Download
- Upload

Doctor

Can:

- View
- Upload
- Write Notes

Cannot delete or edit historical records created by others.

Nursing

Can:

- View
- Upload
- Write Nursing Notes

Operation Theatre

Can:

- View Medical History
- Upload OT Documents

Blood Bank

Can:

- Upload Blood Issue Report
- View relevant medical history

Patient

Can:

- View
- Download their own uploaded reports

Cannot edit or delete records.

Document Viewer

Do not automatically download PDFs.

Instead:

Click PDF



Open built-in viewer.

Only Patients and Super Admin may download files.

Other authorized users may **view only**.

This completes **Part 2**, covering the Laboratory, Radiology, Cashier, and Medical Records (EMR) modules. The next part will continue with Nursing, Admission/IPD, Operation Theatre, Blood Bank, Insurance, Accounts, dashboard improvements, and full inter-module workflow integration.

Go on

HOSPITAL MANAGEMENT SYSTEM MODIFICATION PROMPT (PART 3)

Continue modifying the existing Hospital Management System. Do not rebuild the project. Every module must remain connected through the Patient ID, QR Code, and Medical Record (EMR).

9. NURSING DASHBOARD

The Nursing Dashboard must be redesigned to support clinical care. It should not have billing access.

Dashboard

Display:

- My Assigned Patients
- Today's Admissions
- Today's Discharges

- Pending Nursing Notes
- Search Patient
- QR Code Scanner

Search by:

- Patient ID
- QR Code
- Phone Number
- Patient Name

Nursing Permissions

Nurses can:

- View Patient Profile
- View OPD History
- View Appointment History
- View Admission History
- View Laboratory Reports
- View Radiology Reports
- View Pharmacy History
- View Blood Bank History
- View Surgery History
- View Medical History
- Upload Nursing Notes
- Upload PDF
- Upload Scanned Documents

Nurses cannot:

- Print Bills
- Edit Billing
- Delete Medical Records
- Access Accounts Dashboard
- Edit Insurance Claims

Nursing Notes

Each patient should have:

- Nursing Notes
- Vital Signs
- Progress Notes
- Observation Notes

- Medication Administration Notes
- Upload PDF
- Upload Scanned Nursing Documents

Everything must automatically become part of the patient's Medical Record.

10. ADMISSION (IPD)

Admission should be fully connected with Registration, Billing, Doctor, Nursing, and Medical Records.

Admission Dashboard

Display:

- Search Patient
- Pending Admissions
- Current Admissions
- Bed Allocation
- Ward Information
- Discharge Queue

Search by:

- Patient ID
 - QR Code
 - Phone Number
 - Name
-

Admission Workflow

Doctor recommends admission.

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Admission Department receives the request.

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Patient arrives.

↓

Admission staff assigns:

- Ward
- Room
- Bed
- Admission Date
- Attending Doctor

↓

Generate Admission Bill.

Admission Bill

The bill design must match every other receipt.

Include:

- Phone Number (Compulsory)
- QR Code
- Unique Barcode
- Counter Name
- Counter Staff
- Date
- Time
- Payment Method

Discharge

Discharge must generate a professional discharge summary.

Include:

- Admission Details
- Diagnosis
- Treatment
- Doctor
- Discharge Date
- Uploaded Documents
- Discharge Bill

Automatically save to Medical Record.

11. OPERATION THEATRE (OT)

OT must have its own dashboard.

Dashboard

Display:

- Scheduled Surgeries
 - Completed Surgeries
 - Search Patient
 - QR Scanner
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Doctor Workflow

Doctor searches patient.

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Doctor clicks:

Schedule Operation

Doctor enters:

- Surgery Type
- Priority
- Clinical Notes
- Upload PDF
- Upload Scanned Consent Form
- Planned Date

OT immediately receives the request.

OT Permissions

Operation Theatre staff can:

- View Medical History
- View Laboratory Reports
- View Radiology Reports

- View Blood Bank History
- Upload Surgery Report
- Upload Operative Notes
- Upload PDF
- Upload Scanned Documents

Cannot:

- Edit Billing
- Delete Medical Records

Surgery Completion

After surgery:

Upload:

- Operative Report
- Surgical Notes
- Procedure Summary
- Scanned Documents

Automatically save everything into Medical Records.

12. BLOOD BANK

Blood Bank should work similarly to Laboratory.

Dashboard

Display:

- Pending Requests
- Issued Blood
- Search Patient
- QR Scanner

Blood Request

Doctor requests blood.



Blood Bank receives request.



Blood Bank issues blood.



Upload Issue Report.



Medical Record automatically updates.

Blood Bank Permissions

Can:

- Search Patients
- View Medical History
- View Laboratory Reports
- Upload Blood Issue Report
- Upload PDF
- Write Notes

Cannot:

- Delete Records
- Edit Billing

Blood Bank Bill

Bill must contain:

- Patient Name
- Phone Number (Compulsory)
- QR Code
- Unique Barcode
- Counter Name
- Counter Staff
- Payment Method

- Date
- Time

Same receipt design as all other departments.

13. INSURANCE MODULE

Support Nepal Health Insurance workflow.

Insurance Dashboard

Display:

- Search Patient
- Scan Insurance QR Code
- Pending Claims
- Approved Claims
- Rejected Claims

New Claim

Allow:

- Search Patient
- Scan Insurance QR Code
- Select Insurance Company
- Upload Supporting Documents
- Attach Bills
- Submit Claim

Insurance Types

Support:

- Standard Insurance
 - Staff Discount
 - Poor Patient Scheme
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Staff Discount

Support automatic percentage discount (for example 90% if configured by the hospital).

Poor Patient Card

Allow:

Poor Patient Card Number

Validate card.

Automatically calculate discount.

No manual calculations.

Automatic Billing

If insurance covers the amount:

System automatically reduces payable amount.

Every department should support insurance.

Including:

- Laboratory
 - Radiology
 - Admission
 - OT
 - Blood Bank
 - Pharmacy
-

14. ACCOUNTS DASHBOARD

Accounts staff should only access financial information.

Dashboard includes:

- Revenue

- Refund Requests
 - Discount Requests
 - Pending Approvals
 - Daily Collection
 - Monthly Collection
 - Insurance Payments
-

Search Patient:

By:

- Name
- Phone Number
- Patient ID

Accounts staff cannot edit medical records.

Dashboard Statistics

Large numbers should not display as:

4500000

Instead display:

45 Lakh

125 Lakh

2.5 Crore

Use Nepal numbering style wherever appropriate.

When user clicks the statistic card:

Open a large popup.

Display:

- Exact Amount
- Daily Breakdown
- Monthly Breakdown
- Department Breakdown
- Payment Methods
- Export Options

Do not open another page.

Use responsive modal dialogs.

15. DASHBOARD UI IMPROVEMENTS

Every dashboard should be modern and responsive.

Each statistics card should be clickable.

Examples:

Today's Patients

Today's Revenue

Completed Tasks

Laboratory Reports

Radiology Reports

Insurance Claims

Admissions

Operations

Blood Requests

Clicking a card opens a detailed modal with filters, charts, tables, and complete information without leaving the current page.

End of Part 3.

The final part will cover:

- Complete permission matrix
- Inter-department workflow
- Medical record synchronization
- Patient portal restrictions
- Patient ID card restrictions
- Final UI/UX improvements
- Performance and responsiveness requirements
- Final development rules to ensure every module works together seamlessly.

Go on

HOSPITAL MANAGEMENT SYSTEM MODIFICATION PROMPT (PART 4 – FINAL)

This is the final modification specification. Implement these changes into the existing Hospital Management System. Do not redesign or rebuild working modules. Ensure every module is interconnected, responsive, and follows role-based access control.

16. COMPLETE MODULE CONNECTION (END-TO-END WORKFLOW)

Every department must communicate with each other using the **same Patient ID, Patient QR Code, and Medical Record (EMR)**.

The patient workflow should function as follows:

Patient Registration / Online Appointment

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Registration Counter

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Doctor Consultation

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Doctor decides next action:

- Laboratory Investigation
- Radiology Investigation
- Pharmacy Prescription
- Admission
- Operation Theatre
- Blood Bank
- Follow-up Appointment

↓

The selected department immediately receives the doctor's request.



Department completes its work.



Department uploads reports, scanned documents, or PDF.



Medical Record (EMR) updates automatically.



Doctor immediately sees the updated report.



Patient can also view the report through the Patient Portal.

No manual transfer of reports should ever be required.

17. DOCTOR REQUEST SYSTEM

Replace the basic "Refer" wording with professional medical terminology.

Doctor actions should include:

- Request Laboratory Investigation
- Request Radiology Investigation
- Prescribe Medication
- Request Blood Bank
- Request Admission
- Schedule Surgery
- Schedule Follow-up

Each request must include:

- Doctor Name
- Doctor Department
- Date & Time
- Clinical Notes
- Priority (Routine / Urgent / Emergency)
- Optional PDF Attachment
- Optional Scanned Referral Letter

The receiving department must immediately see this request.

18. MEDICAL RECORD SYNCHRONIZATION

The Medical Record (EMR) must automatically synchronize information from all departments.

Whenever any department uploads a document, it should immediately appear in the patient's timeline.

Supported departments include:

- Doctor Notes
- Nursing Notes
- Laboratory Reports
- Radiology Reports
- Pharmacy Records
- Blood Bank Records
- Admission Records
- Operation Theatre Records
- Insurance Documents

Each record should show:

- Department
- Uploaded By
- Staff Name
- Date
- Time
- File Type
- Note Summary

The timeline should be displayed in chronological order.

19. PATIENT PORTAL PERMISSIONS

Patients should have limited access.

Patients CAN:

- View Appointment History
- View Medical Reports

- View Laboratory Reports
- View Radiology Reports
- View Doctor Notes (if enabled)
- View Medical History
- Download their own uploaded reports and PDFs

Patients CANNOT:

- Download Patient ID Card
- Edit Medical Records
- Edit Personal Clinical History
- Delete Any Record
- View Internal Hospital Notes
- View Financial Administration Data

20. PATIENT ID CARD RESTRICTIONS

This is a mandatory security requirement.

Remove the **Download JPG** option for the Patient ID Card from all locations except:

- Registration Counter
- Super Admin

Patients must **not** be able to download their Patient ID Card.

Doctors must **not** be able to download it.

Nurses must **not** be able to download it.

Laboratory, Radiology, Pharmacy, Accounts, Blood Bank, Admission, and Operation Theatre must **not** have access to download the Patient ID Card.

Only Registration staff and Super Admin may print or regenerate the Patient ID Card.

21. ROLE-BASED ACCESS CONTROL

Super Admin

Full access to all modules.

Can:

- View
 - Create
 - Edit
 - Delete
 - Print
 - Download
 - Manage Users
 - Manage Permissions
 - View Audit Logs
-

Registration

Can:

- Register Patients
- Print Registration Documents
- Print Patient ID Card
- Complete Registration
- Manage Appointment Registration

Cannot modify clinical records.

Doctor

Can:

- Search Patients
- View Complete Patient History
- Create Notes
- Upload PDFs
- Request Laboratory
- Request Radiology
- Schedule Surgery
- Request Admission
- Request Blood Bank
- Prescribe Medication

Cannot:

- Print Bills
 - Edit Financial Records
 - Delete Medical Records
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Nurse

Can:

- View Patient History
- Add Nursing Notes
- Upload Nursing Documents

Cannot modify financial data.

Laboratory

Can:

- View Laboratory Requests
- Upload Laboratory Reports

Cannot edit billing or unrelated medical records.

Radiology

Can:

- View Radiology Requests
- Upload Radiology Reports

Cannot edit billing.

Operation Theatre

Can:

- View Surgery Schedule
- View Medical History
- Upload Operative Reports

Cannot modify billing.

Blood Bank

Can:

- View Blood Requests
- Upload Blood Issue Reports

Cannot edit billing.

Accounts

Can:

- Revenue Reports
- Refund Management
- Discount Management
- Insurance Financial Reports

Cannot modify medical records.

22. RESPONSIVE USER INTERFACE

The entire Hospital Management System must be fully responsive.

It should work correctly on:

- Desktop
- Laptop
- Tablet
- Mobile Phone

All dashboards should resize automatically.

No overlapping buttons or broken layouts.

23. DASHBOARD IMPROVEMENTS

Every dashboard should have:

- Large Search Box
- QR Code Scanner
- Quick Action Buttons
- Summary Cards

- Clickable Statistics
- Professional Icons
- Clean Layout

Clicking a statistics card should open a modal popup with complete details, filters, and export options, without navigating away from the current page.

24. PERFORMANCE REQUIREMENTS

The application should feel fast and interconnected.

- Minimize unnecessary page reloads.
 - Update dashboards in real time where practical.
 - Keep records synchronized across modules.
 - Validate required fields before saving.
 - Prevent duplicate patient records and duplicate requests.
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25. FINAL DEVELOPMENT REQUIREMENTS

- Do **not** remove any existing working functionality unless explicitly requested.
- Keep the current database structure whenever possible and extend it safely.
- Maintain backward compatibility with existing patient records.
- Ensure all modules are fully integrated through the Patient ID, QR Code, and Medical Record (EMR).
- Every receipt across all departments must follow a consistent design and include the patient's **phone number**, **counter name**, and **counter staff name**.
- Every department should use role-based permissions to protect patient privacy and hospital data.
- Every uploaded report or document must automatically become part of the patient's permanent Medical Record.
- Prioritize reliability, security, responsiveness, and a clean user experience throughout the application.

Final Objective

Transform the existing Hospital Management System into a modern, fully integrated hospital information system where all departments—Registration, OPD, Doctor, Laboratory, Radiology, Pharmacy, Nursing, Admission, Operation Theatre, Blood Bank, Insurance, Accounts, Patient Portal, and Super Admin—work together through a single connected patient record, while preserving the existing project and improving its usability, security, and workflow.

